Answers and talk notes

Find Factors and Multiples

What your student is learning

Tonight we find the LEAST COMMON MULTIPLE — the first number two counting patterns land on together.

Answers

Ask your student to explain before you look.

1. **C.** 5, 10, 15, 20. 5, 10, 15, 20 are the first four multiples of 5. Skip count by 5.
2. **A.** 12. 12 is a multiple of both: 3 × 4 = 12 and 4 × 3 = 12.
3. **D.** 15. The LCM of 3 and 5 is 15. It is the first number to show up in both lists.
4. **B.** 18. The LCM of 2 and 9 is 18. It is the first multiple of 9 that is also even.
5. **C.** 30. The LCM of 6 and 10 is 30. Multiples of 6: 6, 12, 18, 24, 30. Multiples of 10: 10, 20, 30.
6. **A.** 36. The LCM of 9 and 12 is 36, so the buses leave together again after 36 minutes.

Talk about it

Which number is the first one in both lists? What does it tell you about when the two patterns meet?

Home mission

Skip-count out loud. Your student says every 3rd number and you say every 4th number. Which number do you both say first? That number is the LCM of 3 and 4.

When they are stuck, ask

- What numbers do you land on skip-counting by this one?
- What is the first number that shows up in both lists?
- Is this question about sharing into groups, or about repeating cycles?

Try not to say

- Multiply the two numbers, that is the LCM.
- Factors and multiples are basically the same.
- The LCM is always the bigger number.